

Heat Loss from Cylinder in Square Enclosure

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1 Problem Statement

Estimate the heat loss from an infinite steel cylinder enclosed in a square plywood box:

- Cylinder diameter: $D = 25 \text{ cm} = 0.25 \text{ m}$
- Cylinder temperature: $T_{cyl} = 90^\circ\text{C}$
- Minimum gap between cylinder and box: $\delta = 10 \text{ cm} = 0.10 \text{ m}$
- Plywood thickness: $t = 1 \text{ cm} = 0.01 \text{ m}$
- Ambient temperature: $T_\infty = 20^\circ\text{C}$
- Air inside enclosure is stagnant (enclosed)
- Calculate heat loss per $L = 30 \text{ cm}$ of length

2 Geometry

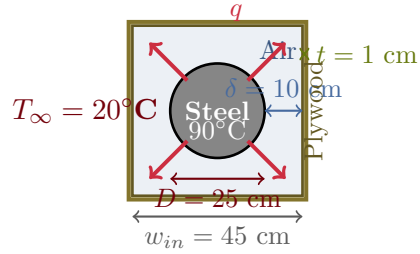


Figure 1: Cross-section of steel cylinder inside square plywood enclosure.

2.1 Calculated Dimensions

$$\text{Cylinder radius: } r_i = D/2 = 0.125 \text{ m} \quad (1)$$

$$\text{Inner box half-width: } a_{in} = r_i + \delta = 0.125 + 0.10 = 0.225 \text{ m} \quad (2)$$

$$\text{Inner box side: } w_{in} = 2 \times a_{in} = 0.45 \text{ m} \quad (3)$$

$$\text{Outer box side: } w_{out} = w_{in} + 2t = 0.47 \text{ m} \quad (4)$$

3 Thermal Resistance Network

Heat flows through three resistances in series:

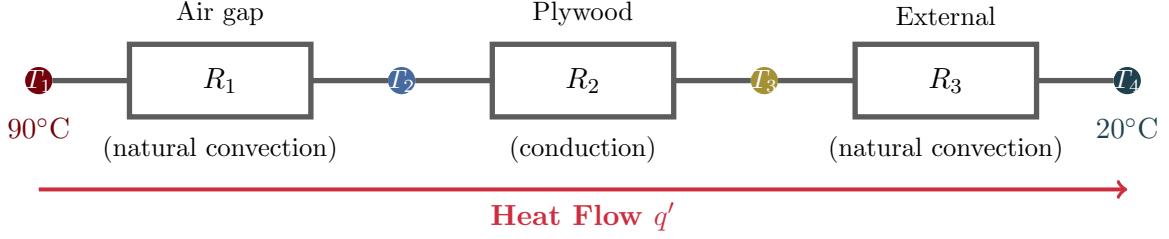


Figure 2: Thermal resistance network (per unit length).

4 Material Properties

4.1 Air at Film Temperature

Estimated film temperature in enclosure: $T_f \approx 55^\circ\text{C} = 328\text{ K}$

Table 1: Air properties at $T_f = 55^\circ\text{C}$

Property	Symbol	Value
Thermal conductivity	k_{air}	$0.0283\text{ W}/(\text{m} \cdot \text{K})$
Kinematic viscosity	ν	$1.82 \times 10^{-5}\text{ m}^2/\text{s}$
Thermal diffusivity	α	$2.59 \times 10^{-5}\text{ m}^2/\text{s}$
Thermal expansion coeff.	β	$3.05 \times 10^{-3}\text{ K}^{-1}$
Prandtl number	Pr	0.703

4.2 Plywood

$$k_{plywood} \approx 0.12\text{ W}/(\text{m} \cdot \text{K}) \quad (5)$$

5 Resistance Calculations (Per Unit Length)

5.1 R_1 : Enclosed Air Gap (Cylinder to Inner Plywood Surface)

For a cylinder inside a square enclosure, the **shape factor** per unit length is:

$$S = \frac{2\pi}{\ln(1.08 \times w_{in}/D)} = \frac{2\pi}{\ln(1.08 \times 0.45/0.25)} = \frac{2\pi}{\ln(1.944)} = \frac{2\pi}{0.665} = 9.45\text{ m}^{-1} \quad (6)$$

For natural convection in the enclosure, we calculate the Rayleigh number using the minimum gap:

$$Ra_\delta = \frac{g\beta\Delta T\delta^3}{\nu\alpha} \quad (7)$$

Estimating $\Delta T \approx 57 \text{ K}$ (between cylinder and inner plywood):

$$Ra_\delta = \frac{9.81 \times 3.05 \times 10^{-3} \times 57 \times (0.10)^3}{1.82 \times 10^{-5} \times 2.59 \times 10^{-5}} = \boxed{3.62 \times 10^6} \quad (8)$$

For enclosed spaces with $Ra > 10^4$, the Nusselt number correlation:

$$Nu = 0.046 \times Ra_\delta^{1/3} = 0.046 \times (3.62 \times 10^6)^{1/3} = 0.046 \times 153 = \boxed{7.0} \quad (9)$$

Effective thermal conductivity:

$$k_{eff} = k_{air} \times Nu = 0.0283 \times 7.0 = 0.198 \text{ W}/(\text{m} \cdot \text{K}) \quad (10)$$

Thermal resistance of air gap:

$$R_1 = \frac{1}{S \times k_{eff}} = \frac{1}{9.45 \times 0.198} = \boxed{0.534 \text{ (m} \cdot \text{K)/W}} \quad (11)$$

5.2 R_2 : Plywood Conduction

For conduction through the plywood wall (approximated as a thin shell):

$$P_{in} = 4 \times w_{in} = 4 \times 0.45 = 1.80 \text{ m} \quad (12)$$

$$R_2 = \frac{t}{k_{plywood} \times P_{in}} = \frac{0.01}{0.12 \times 1.80} = \boxed{0.046 \text{ (m} \cdot \text{K)/W}} \quad (13)$$

5.3 R_3 : External Natural Convection

For natural convection from the outer plywood surface to ambient air:

$$P_{out} = 4 \times w_{out} = 4 \times 0.47 = 1.88 \text{ m} \quad (14)$$

Estimated heat transfer coefficient for natural convection from vertical surfaces:

$$h_{ext} \approx 7 \text{ W}/(\text{m}^2 \cdot \text{K}) \quad (15)$$

$$R_3 = \frac{1}{h_{ext} \times P_{out}} = \frac{1}{7 \times 1.88} = \boxed{0.076 \text{ (m} \cdot \text{K)/W}} \quad (16)$$

6 Total Heat Loss Calculation

6.1 Total Thermal Resistance

$$R_{total} = R_1 + R_2 + R_3 = 0.534 + 0.046 + 0.076 = \boxed{0.656 \text{ (m} \cdot \text{K)/W}} \quad (17)$$

6.2 Heat Loss Per Unit Length

$$q' = \frac{T_{cyl} - T_\infty}{R_{total}} = \frac{90 - 20}{0.656} = \boxed{106.7 \text{ W/m}} \quad (18)$$

6.3 Heat Loss for 30 cm Length

$$Q = q' \times L = 106.7 \times 0.30 = \boxed{32.0 \text{ W}} \quad (19)$$

7 Temperature Distribution

$$T_1 = 90^\circ\text{C} \quad (\text{cylinder surface}) \quad (20)$$

$$T_2 = T_1 - q' \times R_1 = 90 - 106.7 \times 0.534 = \boxed{33.0^\circ\text{C}} \quad (\text{inner plywood}) \quad (21)$$

$$T_3 = T_2 - q' \times R_2 = 33.0 - 106.7 \times 0.046 = \boxed{28.1^\circ\text{C}} \quad (\text{outer plywood}) \quad (22)$$

$$T_4 = 20^\circ\text{C} \quad (\text{ambient}) \quad (23)$$

8 Resistance Breakdown

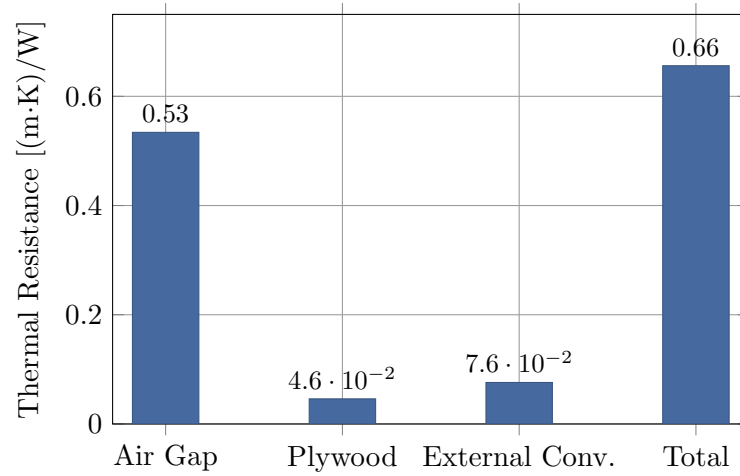


Figure 3: Breakdown of thermal resistances. The enclosed air gap dominates.

9 Summary

Table 2: Heat transfer summary

Parameter	Value
Heat loss per unit length	106.7 W/m
Heat loss for 30 cm	32.0 W
Dominant resistance	Air gap (81.4%)
Plywood resistance	7.0%
External convection	11.6%
Inner plywood temperature	33°C
Outer plywood temperature	28°C

10 Key Observations

1. The **enclosed air gap provides 81% of the total thermal resistance**, making the plywood box an effective insulator despite plywood's relatively high thermal conductivity.
2. The stagnant air acts as insulation, but natural convection ($Nu \approx 7$) still enhances heat transfer by a factor of 7 compared to pure conduction.
3. The plywood contributes only 7% of the resistance—its primary function is containing the air, not insulating.
4. To reduce heat loss further:
 - Increase air gap (but this increases box size and convection effects)
 - Add radiation shields or reflective surfaces
 - Use insulating material instead of air
 - Reduce cylinder temperature